

A Quartic Exponential approximation to the Hypergeometric distribution

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Abstract

The Hypergeometric distribution is a, potentially, asymmetric distribution that, at scale, is typically approximated by the Normal distribution. Numerical differentiation shows that the Quartic Exponential distribution can account for the asymmetry. Such a distribution was constructed using the Stirling approximation, however testing showed that this only approximated the corresponding Hypergeometric distribution to about the same extent as a Normal distribution.

1. Introduction

The Hypergeometric distribution corresponds to sampling without replacement from a finite population of size N where there is a probability of success of $p = K/N = 1 - q$. The proportion of the population that is drawn is $r = n/N = 1 - s$ resulting in x successes. The probability mass function is given as

$$f(x; N, K, n) = \binom{K}{x} \binom{N-K}{n-x} / \binom{N}{n} \equiv \binom{Np}{x} \binom{Nq}{Nr-x} / \binom{N}{Nr} \quad (1)$$

It has the symmetries

$$f(x; N, K, n) = f(n-x; N, N-K, n) \equiv f(Nr-x; N, Nq, Nr) \quad (2)$$

$$f(x; N, K, n) = f(K-x; N, K, N-n) \equiv f(Np-x; N, Np, Ns) \quad (3)$$

$$f(x; N, K, n) = f(x; N, n, K) \equiv f(x; N, Nr, Np) \quad (4)$$

hence

$$x - Npr \rightarrow \pm(x - Npr) \quad (5)$$

$$(p - q)(r - s) \rightarrow \pm(p - q)(r - s) \quad (6)$$

and, as one of an infinite number of examples,

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$$pqrs \rightarrow pqr s \quad (7)$$

The distribution has the mean

$$\mu \equiv \mathbb{E}[X] = n \frac{K}{N} = Npr \quad (8)$$

The median

$$\lfloor n \frac{K}{N} \rfloor \text{ or } \lceil n \frac{K}{N} \rceil \rightarrow Npr \quad (9)$$

The mode

$$\lceil \frac{(n+1)(K+1)}{N+2} - 1 \rceil, \lfloor \frac{(n+1)(K+1)}{N+2} \rfloor \rightarrow Npr \quad (10)$$

The variance

$$\sigma^2 = \mathbb{E}[(X - \mu)^2] = n \cdot \frac{K}{N} \cdot \left(1 - \frac{K}{N}\right) \cdot \frac{N-n}{N-1} \rightarrow Npqrs \quad (11)$$

The skewness

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\mathbb{E}[(X - \mu)^2]^{3/2}} = \frac{(N-2K)\sqrt{N-1} \cdot (N-2n)}{[nK(N-K)(N-n)]^{1/2} \cdot (N-2)} \rightarrow \frac{(p-q)(r-s)}{\sqrt{Npqrs}} \quad (12)$$

The excess kurtosis

$$\gamma_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\mathbb{E}[(X - \mu)^2]^2} - 3 = \frac{(N-1)N^2(N(N+1) - 6K(N-K) - 6n(N-n)) + 6nK(N-K)(N-n)(5N-6)}{nK(N-K)(N-n)(N-2)(N-3)} \rightarrow \frac{1 - 6pq - 6rs + 30pqrs}{Npqrs} \quad (13)$$

At scale this potentially asymmetric distribution is sometimes modelled by the always symmetric Normal distribution. For large N, K and n plotting the derivative of the logarithm of $f(x)$ versus x results in curves which can be fit with a cubic polynomial whose only root in $[0, Nr]$ appears to be at the mode Npr . Note that the Normal distribution would instead result in a straight line.

2. Quartic Exponential distribution

The family of exponential distributions were proposed by Fisher [1], their probability density function can be written as

$$f(x) = \frac{1}{C} \exp(-g(x)) \quad (14)$$

where

$$g(x) = \sum_i \beta_i (x - \mu)^i \quad (15)$$

with the upper bound being an even number and the corresponding β_i being positive. Matz [2] showed that differentiating the integral over the infinite range of $f(x)$ results in

$$\mathbb{E}[(x - \mu)^n] = -\frac{\partial \log C}{\partial \beta_n} \quad (16)$$

For the unimodal symmetric Quartic Exponential distribution O'Toole [3] showed that

$$C = \frac{1}{2} \exp(\theta) K_{1/4}(\theta) \sqrt{\beta_2/\beta_4} \text{ where } \theta = \beta_2^2/(8\beta_4) \quad (17)$$

Combining the two prior equations produces the relationship

$$\beta_2 \mathbb{E}[(x - \mu)^2] + 2\beta_4 \mathbb{E}[(x - \mu)^4] = \frac{1}{2} \Rightarrow \beta_2 \leq \frac{1}{2\sigma^2} \quad (18)$$

the equality of which is obviously satisfied by the Normal distribution.

3. The Stirling approximation

The Stirling approximation was applied to the Hypergeometric distribution, details of the derivation are in the Appendix, with the following result

$$\begin{aligned} \log(f(x)) \approx & -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2}z^2 + \frac{(p-q)(r-s)}{6\sigma}z^3 - \frac{q^3s^3 + q^3r^3 + p^3s^3 + p^3r^3}{12\sigma^2}z^4 \\ & - \frac{(p-q)(r-s)}{2\sigma}z + \frac{p^2r^2 + p^2s^2 + q^2r^2 + q^2s^2}{4\sigma^2}z^2 - \frac{(p^3 - q^3)(r^3 - s^3)}{6\sigma^3}z^3 + \frac{p^4r^4 + p^4s^4 + q^4r^4 + q^4s^4}{8\sigma^4}z^4 \end{aligned} \quad (19)$$

where

$$z = \frac{x - \mu}{\sigma} \quad (20)$$

Note that this does not have a double root at the mode as was suggested by numerical differentiation. It also doesn't satisfy equation 18 although it does satisfy the associated inequality. Finally the normalising value is the same as for the Normal distribution. i.e. Due to the limit of a Bessel function as its argument goes to infinity rather than the finite value dictated by the Quartic Exponential.

None of these arguments, however, mean that the approximation is necessarily bad for which it is required to return to the cubic acquired by differentiating the logarithm of the distribution. It is also sensible to compare with a Normal distribution having the same mean and variance. The plots in figure 1 show that there is seemingly not much difference in accuracy between using the two approximations which have slightly different moments to the Hypergeometric distribution.

4. Discussion

Given that the differential of the logarithm of the Hypergeometric distribution can be fit over much, if not all, of its range by a cubic it was useful to fit a Quartic Exponential distribution to it. Unfortunately nothing seems to be gained by constructing that fit with the Stirling approximation. It is not known whether an analytical solution exists which matches the first four moments of a Hypergeometric distribution with those of a Quartic Exponential distribution. It is always possible to compute the discrete distribution and then use the method of Matz, or some equivalent, to create a continuous distribution which fits.

That raises the question of why bother given the goodness of fit observed for the Normal distribution that matches the first two moments. Given the ease with which it is possible to compute its inverse cumulative distribution function compared to the Hypergeometric or Quartic Exponential this is not a trivial point. The inverse cumulative distribution can be used with a uniform distribution output from a (pseudo) random number generator to perform stochastic simulations so its cost to compute is important.

This is relevant in real world cases where selection without replacement occurs, such as when one chemical reactant chooses another or when an infected person selects several susceptible people to infect. In the standard deterministic models of both processes a bilinear mixing term appears which is the same as the mean of the Hypergeometric distribution and leads to logistic growth to some finite limit. This bilinear mixing term also appears in models of the Kessler syndrome where it leads to hyperbolic growth to infinity after a finite amount of time indicating that those models are too simple to reproduce the limited data that is available.

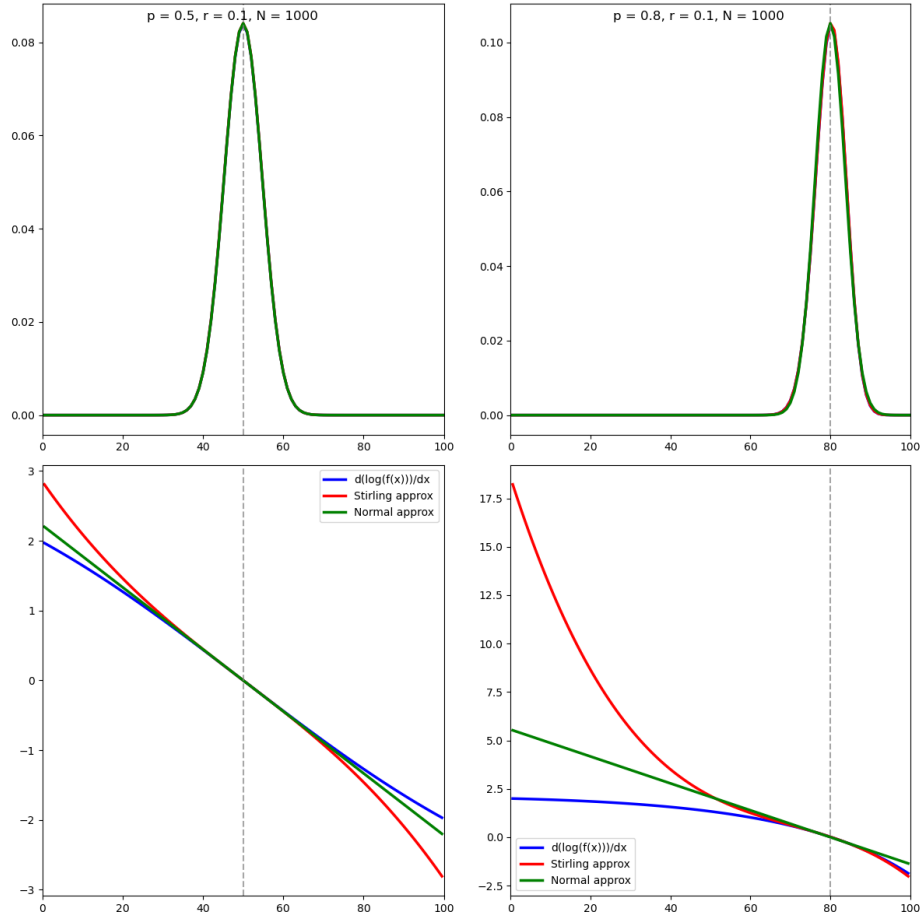


Figure 1. Comparison of approximations

Appendix A. Applying the Stirling approximation to the Hypergeometric distribution

Using the Stirling approximation

$$\log(n!) \approx \frac{1}{2} \log(2\pi n) + n \log(n) - n \quad (\text{A.1})$$

the binomials that form the Hypergeometric distribution can be approximated as

$$\begin{aligned} \log \binom{Np}{x} &\approx Np \log(Np) - x \log(x) - (Np - x) \log(Np - x) \\ &\quad + \frac{1}{2} (\log(Np) - \log(2\pi) - \log(x) - \log(Np - x)) \end{aligned} \quad (\text{A.2})$$

$$\begin{aligned} \log \binom{Nq}{Nr-x} &\approx Nq \log(Nq) - (Nr-x) \log(Nr-x) - (Nq-Nr+x) \log(Nq-Nr+x) \\ &\quad + \frac{1}{2} (\log Nq - \log(2\pi) - \log(Nr-x) - \log(Nq-Nr+x)) \end{aligned} \quad (\text{A.3})$$

$$\begin{aligned} \log \binom{N}{Nr} &\approx N \log(N) - Nr \log(Nr) - Ns \log(Ns) + \frac{1}{2} (\log N - \log(2\pi) - \log(Nr) - \log(Ns)) \\ &= -Nr \log(r) - Ns \log(s) - \frac{1}{2} \log(2\pi Nrs) \end{aligned} \quad (\text{A.4})$$

Substitute

$$x = Npr + \sigma z \quad (\text{A.5})$$

to get

$$\begin{aligned} Np \log(Np) - x \log(x) - (Np-x) \log(Np-x) \\ = -(Npr + \sigma z) \log(r) - (Npr + \sigma z) \log\left(1 + \frac{\sigma z}{Npr}\right) \\ - (Nps - \sigma z) \log(s) - (Nps - \sigma z) \log\left(1 - \frac{\sigma z}{Nps}\right) \end{aligned} \quad (\text{A.6})$$

$$\begin{aligned} \log(Np) - \log(2\pi) - \log(x) - \log(Np-x) \\ = -\log(2\pi Nprs) - \log\left(1 + \frac{\sigma z}{Npr}\right) - \log\left(1 - \frac{\sigma z}{Nps}\right) \end{aligned} \quad (\text{A.7})$$

$$\begin{aligned} Nq \log(Nq) - (Nr-x) \log(Nr-x) - (Nq-Nr+x) \log(Nq-Nr+x) \\ = -(Nqr - \sigma z) \log(r) - (Nqr - \sigma z) \log\left(1 - \frac{\sigma z}{Nqr}\right) \\ - (Nqs + \sigma z) \log(s) - (Nqs + \sigma z) \log\left(1 + \frac{\sigma z}{Nqs}\right) \end{aligned} \quad (\text{A.8})$$

$$\begin{aligned} \log Nq - \log(2\pi) - \log(Nr-x) - \log(Nq-Nr+x) \\ = -\log(2\pi Nqrs) - \log\left(1 - \frac{\sigma z}{Nqr}\right) - \log\left(1 + \frac{\sigma z}{Nqs}\right) \end{aligned} \quad (\text{A.9})$$

Combining

$$Nr \log(r) + Ns \log(s)$$

$$\begin{aligned} & - (Npr + \sigma z) \log(r) - (Npr + \sigma z) \log\left(1 + \frac{\sigma z}{Npr}\right) \\ & - (Nps - \sigma z) \log(s) - (Nps - \sigma z) \log\left(1 - \frac{\sigma z}{Nps}\right) \\ & - (Nqr - \sigma z) \log(r) - (Nqr - \sigma z) \log\left(1 - \frac{\sigma z}{Nqr}\right) \\ & - (Nqs + \sigma z) \log(s) - (Nqs + \sigma z) \log\left(1 + \frac{\sigma z}{Nqs}\right) \\ & = - (Npr + \sigma z) \log\left(1 + \frac{\sigma z}{Npr}\right) - (Nps - \sigma z) \log\left(1 - \frac{\sigma z}{Nps}\right) \\ & - (Nqr - \sigma z) \log\left(1 - \frac{\sigma z}{Nqr}\right) - (Nqs + \sigma z) \log\left(1 + \frac{\sigma z}{Nqs}\right) \end{aligned} \quad (\text{A.10})$$

and

$$\log(2\pi Nrs)$$

$$\begin{aligned} & - \log(2\pi Nprs) - \log\left(1 + \frac{\sigma z}{Npr}\right) - \log\left(1 - \frac{\sigma z}{Nps}\right) \\ & - \log(2\pi Nqrs) - \log\left(1 - \frac{\sigma z}{Nqr}\right) - \log\left(1 + \frac{\sigma z}{Nqs}\right) \\ & = - \log(2\pi Npqrs) - \log\left(1 + \frac{\sigma z}{Npr}\right) - \log\left(1 - \frac{\sigma z}{Nps}\right) - \log\left(1 - \frac{\sigma z}{Nqr}\right) - \log\left(1 + \frac{\sigma z}{Nqs}\right) \end{aligned} \quad (\text{A.11})$$

Expanding the logarithms as Taylor series up to the quartic results in

$$\begin{aligned} & - (Npr + \sigma z) \log\left(1 + \frac{\sigma z}{Npr}\right) - (Nps - \sigma z) \log\left(1 - \frac{\sigma z}{Nps}\right) \\ & - (Nqr - \sigma z) \log\left(1 - \frac{\sigma z}{Nqr}\right) - (Nqs + \sigma z) \log\left(1 + \frac{\sigma z}{Nqs}\right) \\ & \approx -\frac{1}{2}z^2 + \frac{(p-q)(r-s)}{6\sigma}z^3 - \frac{q^3s^3 + q^3r^3 + p^3s^3 + p^3r^3}{12\sigma^2}z^4 + \dots \end{aligned} \quad (\text{A.12})$$

and

$$\begin{aligned} & \log\left(1 + \frac{\sigma z}{Npr}\right) + \log\left(1 - \frac{\sigma z}{Nps}\right) + \log\left(1 - \frac{\sigma z}{Nqr}\right) + \log\left(1 + \frac{\sigma z}{Nqs}\right) \\ & \approx \frac{(p-q)(r-s)}{\sigma}z - \frac{p^2r^2 + p^2s^2 + q^2r^2 + q^2s^2}{2\sigma^2}z^2 + \frac{(p^3 - q^3)(r^3 - s^3)}{3\sigma^3}z^3 - \frac{p^4r^4 + p^4s^4 + q^4r^4 + q^4s^4}{4\sigma^4}z^4 + \dots \end{aligned} \quad (\text{A.13})$$

Gathering together

$$\begin{aligned} \log(f(x)) \approx & -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2}z^2 + \frac{(p-q)(r-s)}{6\sigma}z^3 - \frac{q^3s^3 + q^3r^3 + p^3s^3 + p^3r^3}{12\sigma^2}z^4 \\ & - \frac{(p-q)(r-s)}{2\sigma}z + \frac{p^2r^2 + p^2s^2 + q^2r^2 + q^2s^2}{4\sigma^2}z^2 - \frac{(p^3 - q^3)(r^3 - s^3)}{6\sigma^3}z^3 + \frac{p^4r^4 + p^4s^4 + q^4r^4 + q^4s^4}{8\sigma^4}z^4 \quad (\text{A.14}) \end{aligned}$$

References

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